THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET A

Chapters 13 & 14: **Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Think of a number n. Add 7, multiply by 3, then subtract 21. The result is: 1

(a) $3n + 7$ **(b)** $n + 7$ **(c)** $7n$ **(d)** $3n$

2. The two-digit number with tens digit 3 and ones digit 7 can be written as: 1

(a) $3 + 7$ **(b)** $10 \times 3 + 7$ **(c)** $10 \times 7 + 3$ **(d)** 3×7

3. The sum of 35 and its reverse 53 is: 1

(a) 88 **(b)** 99 **(c)** 72 **(d)** 8

4. In a number pyramid, each block is the sum of the two below it. The bottom row is 3, 5, 6. The top block is: 1

(a) 14 **(b)** 21 **(c)** 24 **(d)** 19

5. The expression $3(n + 4) - 2n$ simplifies to: 1

(a) $5n + 12$ **(b)** $n + 12$ **(c)** $n + 4$ **(d)** $n + 7$

6. The area of a rectangle of length 24 cm and breadth 5 cm is: 1

(a) 125 cm² **(b)** 58 cm² **(c)** 29 cm² **(d)** 120 cm²

7. The area of a triangle with base 20 cm and height 7 cm is: 1

(a) 27 cm² **(b)** 90 cm² **(c)** 140 cm² **(d)** 70 cm²

8. The area of a parallelogram with base 15 cm and height 8 cm is: 1

(a) 120 cm² **(b)** 46 cm² **(c)** 128 cm² **(d)** 60 cm²

9. The area of a trapezium with parallel sides 8 cm and 12 cm and height 5 cm is: 1

(a) 50 cm² **(b)** 100 cm² **(c)** 58 cm² **(d)** 96 cm²

10. The area of a rhombus with diagonals 10 cm and 8 cm is: 1

(a) 40 cm² **(b)** 20 cm² **(c)** 18 cm² **(d)** 80 cm²

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why is $n(n + 1)$ always even? 2

12. Explain why a number is divisible by 3 if the sum of its digits is. 2

13. Convert 3.5 m² to cm². 2

14. Find the area of a square of perimeter 52 cm. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Two numbers add to 18. Find the numbers that give the largest product and the product. 3

16. A triangular field has base 18 m and height 12 m. Find its area and the cost of levelling at ₹ 12 per m². 3

17. Two numbers add to 22. Find the numbers that give the largest product and the product. 3

OR

Two numbers add to 30. Find the numbers that give the largest product and the product. 3

18. A triangular field has base 25 m and height 14 m. Find its area and the cost of levelling at ₹ 12 per m². 3

OR

A triangular field has base 30 m and height 8 m. Find its area and the cost of levelling at ₹ 12 per m². 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Show that $n^2 - 1$ is divisible by 8 for any odd n. 5
(ii) Check for $n = 9$.

OR

(i) Explain how grids of numbers (e.g. 3×3 blocks) have a sum equal to 9 times the centre. 5
(ii) Verify with centre 15.

20. (i) Find the area of a parallelogram with base 24 cm and height 15 cm. 5
(ii) If its slant side is 17 cm, find the area of the triangle cut off by the height.

OR

(i) Rectangles of equal perimeter 40 cm can have different areas. Give four examples. 5
(ii) Which is largest?

END OF QUESTION PAPER — Total: 40 Marks

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